

# Touhidul Alam Seyam

## MACHINE LEARNING ENGINEER | AI AND AGENTIC SYSTEMS

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## RESEARCH PROFILE

Applied machine-learning researcher and software engineer working across computer vision, tabular modeling, healthcare analytics, agricultural AI, LLM inference, and high-performance computing. Author or co-author of ten source-backed research works, including two IEEE COMPAS papers, two SpringerOpen journal articles, and a 2026 Springer conference chapter. Builds reproducible Python pipelines and production-facing web systems that translate experiments into usable tools.

## RESEARCH AND ENGINEERING SKILLS

**Machine Learning:** scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, CNNs, transfer learning, SVM, clustering, feature engineering, cross-validation, model evaluation

**Engineering:** Python, TypeScript, Next.js, React, Node.js, FastAPI, Django, PostgreSQL, MongoDB, Redis, Docker, Git, Zig, Bun

**Research Practice:** reproducible experiments, fold-safe validation, benchmark design, data pipelines, academic writing, literature review, publication preparation

## ENGINEERING EXPERIENCE

### Software Engineer | Agentic Institute | Remote | Dec 2025-Present

- Contribute to application development and AI-enabled, agentic software systems.
- Work across implementation, integration, testing, and delivery using modern web and backend technologies.

### Software Engineer | Hello World Communications Ltd | Chattogram, Bangladesh | Aug 2024-Present

- Develop and deploy scalable applications and AI-enabled features using TypeScript, React, Next.js, Node.js, Python, and PostgreSQL.
- Bridge experimental software and production application requirements through code review, system design, deployment, and support.

## SELECTED RESEARCH PROJECTS

### Smartphone Addiction Prediction | Python, LightGBM, XGBoost, CatBoost |

[github.com/Seyamalam/playground-series-s6e8](https://github.com/Seyamalam/playground-series-s6e8)

- Built leakage-safe, five-fold out-of-fold pipelines with nested target encoding, matched-fold experiment screening, schema checks, and rank blending.

### Kaggriculture Autonomous Agent | Python, Simulation, Evaluation |

[github.com/Seyamalam/Kaggriculture](https://github.com/Seyamalam/Kaggriculture)

- Engineered a deterministic farming agent and reproducible evaluation harness with seat-swapped tournaments, frozen replay corpora, manifest hashing, and regression gates.

### bun-scikit | TypeScript, Bun, Zig | [github.com/Seyamalam/bun-scikit](https://github.com/Seyamalam/bun-scikit)

- Implemented a scikit-learn-inspired API with native Zig acceleration, parity contracts, benchmarks, documentation coverage checks, and broad model/preprocessing/evaluation support.

## PUBLICATIONS

- “Architectures for AI-Driven Visual Assistance: Evaluating Server-Mediated Mobile and Direct Access Desktop Client Implementations.” Lecture Notes in Networks and Systems, 2026. DOI: 10.1007/978-3-032-15764-5\_45
- “Enhancing Agricultural Diagnostics: Advanced Training of Pre-Trained CNN Models for Paddy Leaf Disease Detection.” Machine Learning Research, 2025. DOI: 10.11648/j.ml.20251001.11
- “Next-Generation K-Means Clustering: Mojo-Driven Performance for Big Data.” International Journal of Intelligent Information Systems, 2025. DOI: 10.11648/j.ijis.20251401.12
- “Efficient Malware Classification Using Multiprocessing and Bag-of-Words Vectorization.” Advances in Networks, 2025. DOI: 10.11648/j.net.20251201.12
- “Fine-tuning LLaMA 2 interference: a comparative study of language implementations for optimal efficiency.” arXiv, 2025. DOI: 10.48550/arXiv.2502.01651
- “AgriScan: Next.js powered cross-platform solution for automated plant disease diagnosis and crop health management.” Journal of Electrical Systems and Information Technology, 2024. DOI: 10.1186/s43067-024-00169-7
- “Application of Machine Learning K-Means Clustering and Linear Regression in Determining the Risk Level of Pulmonary Tuberculosis.” IEEE COMPAS, 2024. DOI: 10.1109/COMPAS60761.2024.10796963
- “Enhancing Cardiovascular Risk Prediction Using Support Vector Machines and Advanced Machine Learning Algorithms.” IEEE COMPAS, 2024. DOI: 10.1109/COMPAS60761.2024.10796805
- “Comparing pre-trained models for efficient leaf disease detection: a study on custom CNN.” Journal of Electrical Systems and Information Technology, 2024. DOI: 10.1186/s43067-024-00137-1
- “Comparative Performance Evaluation of Classical Machine Learning and Quantum SVM for Heart Disease Prediction using a Quantum-Featured Dataset.” Sonargaon University Journal, 2025. Link: [su.edu.bd/web\\_assets/journal/journal\\_five/journal3.pdf](http://su.edu.bd/web_assets/journal/journal_five/journal3.pdf)

## EDUCATION

**B.Sc. (Hons.) in Computer Science and Engineering | BGC Trust University Bangladesh | Chattogram, Bangladesh | Expected Dec 2026**

Currently in the 8th semester. Research interests: applied machine learning, computer vision, agricultural AI, healthcare analytics, LLM systems, and agentic systems.

## RESEARCH PROFILES

ORCID: 0009-0007-7512-1893 | Google Scholar: [scholar.google.com/citations?user=gRkTVYEAAAAJ](https://scholar.google.com/citations?user=gRkTVYEAAAAJ) | GitHub: [github.com/Seyamalam](https://github.com/Seyamalam)